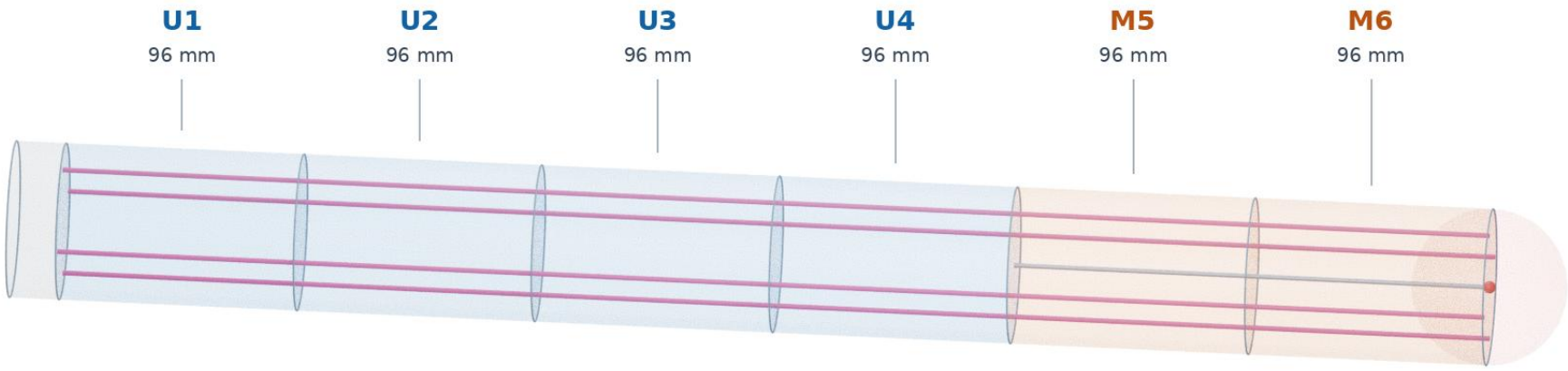
U4 + M2 | Exact simulation reference

500 g nominal v3 · source revision bee65b7 · model coordinates, not a manufacturing release

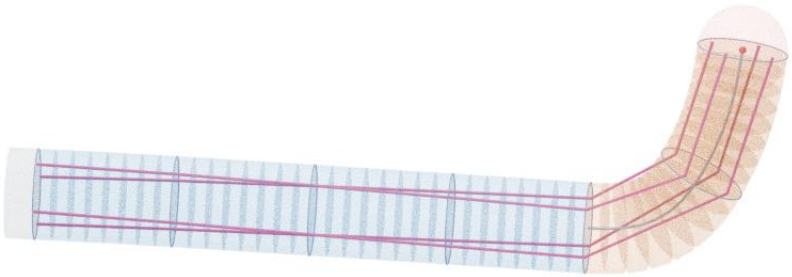


20 mm rigid interface + 576 mm flexible front = 596 mm clamp-to-head

All tendon centers: radius 24 mm. Same-direction A/B/C routes coincide; visible paths show C.

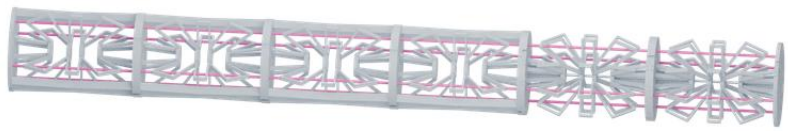
Saved 500 g equilibrium

Sample 386 · yaw 90.00° · elevation 20.02°



Original CAD reference

Original meshes retained unchanged; not conformed.



Required model dimensions

Backbone (each M)	**Ø1.8 × 96 mm**
Anchors from clamp	**404 / 500 / 596 mm**
Head load	**500 g point at X=596 mm**
Tendon model	**12 × Ø2 mm; 1 m free rope**
Latest tension control	**3 N target / 2.5 N floor**

PARAMETERS MATCH THE SOLVER. PHYSICAL CONFORMANCE IS NOT ESTABLISHED.

1. Three finite-diameter ropes cannot occupy the same route through the U section.
2. Original mesh cylindrical envelope ≈ Ø63.311 mm; simulation envelope = Ø63 mm.
3. Backbone restraint and module continuity are ideal constraints; no extra fixtures are implied.

Envelope surfaces are analytical visualization. Source CAD is not silently scaled; stored results are not rerun.